Sungjun Han

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Education

Imperial College London *2019.9 ~ Present*	Electronic and Information Engineering Meng (*Hons*) ***1st year Upper Second Degree*** *Including Programming for Engineers, Digital and Computer Architecture, and Analysis and Design of Circuits* ***2nd year Lower Second Degree*** *Including Instruction Architecture and Compilers, Discrete Mathematics, Software Systems, Information Processing, and Control Systems* ***3rd year Predicted Upper Second Degree*** *Including Machine Learning, Deep Learning, Advanced Databases, Distributed Algorithms, and System Performance Engineering*
Seoul Foreign School *2013.06 ~ 2019.06*	International Baccalaureate Diploma of Higher Education 41/45 *Higher Level: Mathematics 7, Physics 7, Economics 6* *Standard Level: Chemistry 6, English-A 6, Korean-A 6*

Professional Work Experience

Chain Lightning Company *2021.07 ~ 2021.09*	Startup Intern	• Assisted in the design and coding of a mobile platform for payments of electrical vehicle charging in Korea. • Participated in the development of the back end of the mobile app using Django Framework via Python. • Coded the log-in and sign-up section of the app, including features such as email verification, email and password security check, and social logins.
SOR Lab, Seoul National University *2020.07 ~ 2020.09*	Summer Research Intern	• Security Optimization Research lab of Seoul National University Department of Electrical and Computer Engineering. • Research in malware detection through the usage of AI. • Participated in research on dynamic detection of malware application upon instruction stage in virtual machine. • Coded a Python algorithm using Intel Pin to extract the instruction set of the active malware.

Projects

Mars Rover *Group Project* *2021.05 ~ 2021.06*	• Designed an autonomous rover system functional in a remote location without direct supervision. • Worked on the Server – Rover connection portion of the rover, connecting the user to the rover by the ESP-32 Wi-Fi attached to the Arduino UNO board. • Developed a TCP/IP server communication system which sent out instructions guiding the motion of the rover and received data regarding the remaining battery power and the distance and color of objects in its vision.
C90 Compiler *Group Project* *2021.01 ~ 2021.03*	• Designed and coded an ANSI C to MIPS compiler capable of functions such as local variables, arithmetic and logical expressions, if-then-else statements, while-loops, for-loops, recursive function calls, and more. • Implemented the lexical analyzer and regular expressions using Flex, parser using Yacc, and the abstract syntax tree using C++.
MIPS CPU *Group Project* *2020.11 ~ 2020.12*	• Developed a system Verilog implementation of a five-stage pipelined, MIPS processor including the fetch, decode, execute, memory, and writeback stage. • Worked as a part of the test bench team to create R-type, J-type, and I-type instruction samples to test the functionality of the bus CPUs.
Automated Detection of Diabetic Retinopathy *Personal Project* *2016.09 ~ 2017.09*	• Automated the detection of diabetic retinopathy from a retinal photograph. • Employed Mathematica to apply 3D regression on RGB data of retinal images to detect lesions of diabetic retinopathy with a 92.3% success rate. • Received 2nd place award in Korea's KSEF science fair, participated in European Commission initiated conference EUCYS with personal booth.

Extracurriculars

Imperial College Data Science Society (ICDSS) *2021.9 ~ Present*	• Imperial College society promoting data science awareness amongst the student body. • Attended weekly lectures regarding the application of Python, TensorFlow and Matplotlib to the field of Data Science.
Imperial College Korean Society (ICKS) *2019.9 ~ Present*	• Imperial College society for the networking and socializing of Korean students. • Worked as an event officer of the society to plan and execute events such as the Freshers Welcome Fair, Orientation, and overnight networking events. • Captain of the ICKS Football team, planned practices and friendlies with other university football teams of London.